A Scalable Aggregating Algorithm for Races

Peter Cotton*

November 2026

Abstract

Vovk's Aggregating Algorithm predicts with a substitution of its weighted mixture of experts. For the log-loss of a contest winner under a Thurstone race model, the mixture of races over Gaussian uncertainty in the abilities is again a race, with that uncertainty added to the performance covariance. The algorithm's prediction under a Gaussian posterior is therefore one forward pass, linear in the field size when the covariance has factor structure, where the softmax analogue needs Monte Carlo. We state the fact and check it numerically. Online, the mixed prediction was no more accurate than the plug-in race at the posterior mean, even with the exact posterior; what it supplies is the algorithm's worst-case guarantee at the cost of one race.

1 Setting

A contest has n entrants with abilities $x \in \mathbb{R}^n$. Entrant i performs $Y_i = x_i + \varepsilon_i$ with $\varepsilon \sim N(0, \Sigma)$, and the lowest performance wins. Write $p_\Sigma(x)$ for the vector of win probabilities and

$$\ell(x, y) = -\ln p_{\Sigma,y}(x)$$

for the log-loss of an observed winner y. When $\Sigma = VV^\top + \operatorname{diag}(D)$ with V of low rank, $p_\Sigma(x)$ costs time linear in n, and so does its inverse p_Σ^{-1}, which returns the abilities (up to translation) that give a target vector of win probabilities (Cotton, 2026). The race map is onto the open simplex, so the inverse is defined everywhere.

An online learner sees contests $t = 1, \dots, T$, each a field $F_t \subseteq \{1, \dots, N\}$ from a population of N entrants, and predicts a distribution over the winner before y_t is revealed.

2 The Aggregating Algorithm and its substitution

The Aggregating Algorithm (Vovk, 1990, 2001) keeps weights P_t over experts, here ability vectors x, updated as $P_{t+1}(x) \propto P_t(x)\, e^{-\eta\, \ell(x_{F_t}, y_t)}$, and predicts with a substitution $s(P_t)$. The loss is η-mixable if some s satisfies

$$\ell(s(P), y) \le -\tfrac{1}{\eta} \ln \mathbb{E}_{x \sim P}\, e^{-\eta\, \ell(x,y)} \qquad \text{for all } P,\ y, \tag{1}$$

and then the cumulative loss is at most $-\frac{1}{\eta} \ln \mathbb{E}_{x \sim P_1} e^{-\eta L_T(x)}$, where $L_T(x)$ is expert x's cumulative loss. Against any comparator distribution Q this is at most $\mathbb{E}_Q L_T + \frac{1}{\eta} \operatorname{KL}(Q \| P_1)$.

*peter.cotton@microprediction.com. The seeded scripts behind every number are in `papers/aggregating/` in `github.com/microprediction/winning`.

At $\eta = 1$ the right side of (1) is $-\ln \mathbb{E}_P\, p_{\Sigma,y}(x)$, so

$$s(P) = p_\Sigma^{-1}\big(\mathbb{E}_{x\sim P}\, p_\Sigma(x)\big) \tag{2}$$

meets (1) with equality for every y. This is not new. Log-loss is 1-mixable whatever the parameterisation of the forecast, provided the parameterisation reaches every interior forecast (van Erven et al., 2012, Section 6), and Foster et al. (2018) give the softmax case, $s(\pi) = \sigma^{-1}(\mathbb{E}_\pi \sigma(x))$, as the route to logarithmic regret for multiclass logistic regression. The Thurstone case needs only that p_Σ is invertible, and (2) is computable with the inverse above.

3 Gaussian closure

The step that is specific to races is the expectation in (2).

Proposition 1. *If $x \sim N(m, S)$ then $\mathbb{E}\, p_\Sigma(x) = p_{\Sigma+S}(m)$.*

Proof. $Y = x + \varepsilon$ with independent $x \sim N(m, S)$ and $\varepsilon \sim N(0, \Sigma)$ is $N(m, \Sigma + S)$, and averaging the event $\{Y_y < Y_j \ \forall j \neq y\}$ over x gives its probability under that law. □

This is the multinomial-probit version of the familiar $\mathbb{E}\,\Phi(w^\top z) = \Phi(m^\top z/\sqrt{1 + z^\top S z})$ (Rasmussen and Williams, 2006, Section 3.9) and of random-coefficient probit (Hausman and Wise, 1978; Train, 2009). Its consequence for aggregation is that under a Gaussian posterior the algorithm's prediction is a single race, with performance covariance $\Sigma + S$. If S is itself low rank plus diagonal, as a factor posterior is, $\Sigma + S$ keeps that form and the prediction costs one forward pass, linear in n. Passing it through p_Σ^{-1} expresses it, if wanted, as abilities in the original model.

The softmax has no such closure. The fast mixable method of Jézéquel et al. (2021) runs the Aggregating Algorithm on quadratic surrogates, so its posterior is exactly Gaussian, and then estimates $\mathbb{E}\,\sigma(x)$ by sampling; they used 100 draws. For races the same step is exact.

The closure also covers whole orderings. A Gaussian mixture of Thurstone ranking laws with covariance Σ is the Thurstone ranking law with covariance $\Sigma + S$, so the ranking log-loss is 1-mixable with a substitution inside the general-covariance Thurstone family. Plackett–Luce is not closed in this way: mixtures of Plackett–Luce laws violate independence of irrelevant alternatives (Train, 2009) and form a strictly larger class (Zhao et al., 2016). Since a substitution meeting (1) at $\eta = 1$ for every ordering must equal the mixture, the ranking log-loss with a Plackett–Luce substitution is not 1-mixable. The closure needs the general covariance: a Case V model (independent, equal-variance noise) is not closed, because S adds correlation.

4 What the guarantee covers

The exact aggregating posterior under probit likelihoods is not Gaussian; it is unified skew-normal (Durante, 2019). The one-pass prediction therefore applies exactly to a Gaussian posterior: the surrogate posterior of Jézéquel et al. (2021), or a Laplace or expectation-propagation approximation, which carries no regret guarantee of its own. For the exact Bayesian mixture with a Gaussian prior, regret bounds that need only bounded curvature of the log-likelihood (Kakade and Ng, 2005; Kakade et al., 2005) apply to probit. Whether improperness is needed at all for probit is less clear than for the logistic, whose proper lower bound comes from its exponential tail. We claim the computation, not a new rate, and the checks below find no accuracy gain from it in ordinary rating streams.

5 Checks

All numbers come from `checks.py` and `exact_bayes.py` with fixed seeds.

Substitution. For random mixtures P of 2 to 50 ability vectors in fields of 3 to 25, deliberately non-Gaussian, (2) computed with the race inverse meets (1) with equality to within 8.5×10^{-9} in every coordinate, the inverse's own tolerance.

Closure. For posteriors $N(m, LL^\top + \operatorname{diag} d)$ in fields of 4 to 12 with L of rank 1 or 2, the single race at covariance $I + LL^\top + \operatorname{diag} d$ agrees with 4×10^6 simulated contests to within 1.7 standard errors in every entry, at 1 to 20 ms per call.

Scale. For factor posteriors over 1,000 entrants (rank 2) the mixed race takes 0.70 s. Averaging 100 plug-in races over posterior draws, the Monte Carlo step of Jézéquel et al. (2021), takes 0.74 s and misses the ten favourites' probabilities by up to 25% (median over five repetitions); reaching 1% would need some 6×10^4 draws. At 5,000 entrants and rank 3 the mixed race takes 53 s, against 3.6 s for 100 draws that miss by up to 53%.

Online. Entrants with standard normal abilities race in fields of 5 drawn from a population of N, for $T = 1000$ contests, five seeds. A Gaussian posterior is updated by one Laplace step per contest on the field's block, and predicts either by the mixed race $p_{I+S_{FF}}(m_F)$ or by the plug-in race $p_I(m_F)$. Summed over the 1000 contests, the mixed race lost more than the plug-in by 1.0 ± 0.6 nats at $N = 20$ and 3.9 ± 2.0 nats at $N = 400$, where each entrant has about a dozen contests. A diagonal posterior did no better. To separate the substitution from the Laplace approximation, a small case ($N = 6$, fields of 3, 40 contests, 30 seeds) was run with the exact posterior on 6×10^4 prior particles. Exact Bayes, the Aggregating Algorithm itself, and the plug-in race at the exact posterior mean differed by 0.007 ± 0.118 nats, which is nothing; the Laplace mixed race lost 0.23 ± 0.09 nats to it, the Laplace plug-in 0.02 ± 0.03. In these tests the mixture's smoothing is second order, and a crude Gaussian posterior overstates the variance it smooths with.

6 Remarks

Glicko's expected score shrinks the rating difference by $g(\sqrt{RD_i^2 + RD_j^2})$ (Glickman, 1999), and TrueSkill adds the players' variances to the performance variance (Herbrich et al., 2006). Both are approximate Gaussian aggregating predictions for two players. Proposition 1 gives the exact version for any number of entrants with correlated performances and correlated uncertainty, at the cost of one race.

References

P. Cotton. Scalable inversion of contests with correlated performances, including softmax and multinomial probit. arXiv:2609.01133, 2026.

D. Durante. Conjugate Bayes for probit regression via unified skew-normal distributions. *Biometrika*, 106(4):765–779, 2019.

D. J. Foster, S. Kale, H. Luo, M. Mohri, and K. Sridharan. Logistic regression: the importance of being improper. In *Proc. COLT*, PMLR 75, 2018.

M. E. Glickman. Parameter estimation in large dynamic paired comparison experiments. *JRSS C*, 48(3):377–394, 1999.

J. A. Hausman and D. A. Wise. A conditional probit model for qualitative choice. *Econometrica*, 46(2):403–426, 1978.

R. Herbrich, T. Minka, and T. Graepel. TrueSkill: a Bayesian skill rating system. In *NIPS 19*, 2006.

R. Jézéquel, P. Gaillard, and A. Rudi. Mixability made efficient: fast online multiclass logistic regression. In *NeurIPS 34*, 2021.

S. M. Kakade and A. Y. Ng. Online bounds for Bayesian algorithms. In *NIPS 17*, 2005.

S. M. Kakade, M. W. Seeger, and D. P. Foster. Worst-case bounds for Gaussian process models. In *NIPS 18*, 2005.

C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.

K. E. Train. *Discrete Choice Methods with Simulation*. Cambridge University Press, 2nd edition, 2009.

T. van Erven, M. D. Reid, and R. C. Williamson. Mixability is Bayes risk curvature relative to log loss. *JMLR*, 13:1639–1663, 2012.

V. Vovk. Aggregating strategies. In *Proc. COLT*, 371–386, 1990.

V. Vovk. Competitive on-line statistics. *International Statistical Review*, 69(2):213–248, 2001.

Z. Zhao, P. Piech, and L. Xia. Learning mixtures of Plackett–Luce models. In *Proc. ICML*, PMLR 48, 2016.